



DESIGN REPORT

Of

Laboratory Information System (LIS) for

Medical Laboratories with Semi-Automatic

Equipments

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Chapter 1: Description of the New System

1.1 Narrative

The suggested Laboratory Information System (LIS) is an internet-based system designed to streamline and automate the day-to-day processes of small and medium-sized medical labs working with semi-automatic lab equipment. The system eliminates the current paper and Excel-based process management and introduces a secure, centralized system for managing the entire laboratory operation process, from patients' booking to report delivering.

The process starts with the online booking process, when patients make their appointments to perform laboratory tests via the web interface choosing necessary test, preferable date and time. All the appointment information is stored in the centralized PostgreSQL database.

The registration of the client at the laboratory takes place through the receptionist, where the system will register either existing or new clients depending on the presence of their patient record in the database, either by using their identification numbers or other related details. The system automatically creates a unique record and barcode for the specimen collected. The barcode is used to mark the collected specimen to facilitate its identification.

Lab technicians collect the labeled specimens and conduct the necessary test using semi-automatic laboratory equipment. Since the equipment cannot integrate with the system, the technicians manually fill the system using digital forms. Validation rules are used to prevent typing mistakes in the system.

When the results have been entered, the system will create professional laboratory reports automatically using the predefined report templates. All the reports will be created following the standardized form which has been followed due to the inconsistencies that were created by the use of various Excel templates. The generated reports will be securely stored in the centralized database to facilitate retrieval whenever required by the authorized user.

The billing functionality has been incorporated into the patient management process. The system will automatically generate the invoices, calculate the amount payable for the selected laboratory test, record the payments made, and maintain the billing history without the need to use any other software.

The notification module notifies the patients of the availability of the laboratory reports. Based on the configuration of the laboratory, patients will be notified via SMS or Email without revealing any confidential information. The patients will collect the printed report from the laboratory or access the reports online via the patient portal.

This system adopts a Role Based Access Control approach, whereby every user can access only the functions assigned according to his/her duties. Receptionists will be responsible for managing appointments and patient registration, lab technicians will perform sample testing and results

reporting, doctors or medical officers authorized by them will confirm reports where required, administrators will have the authority to manage users and system parameters whereas patients will have access only to their appointment and lab reports.

For ensuring data security and reliability, a centralized database in PostgreSQL format is used. Communication between the client and the server is secured through HTTPS whereas authentication of the user takes place through JWT. Every significant activity performed within the system is logged into audit logs.

The system uses three layers including the React.js interface, Node.js application layer and PostgreSQL database. React.js allows the user to create an interactive and intuitive interface which will work well on desktops, tablets, and smartphones. Business logic, request processing, authentication, and interaction with the database is carried out by the Node.js backend. All patient data, test results, appointment details, billing data and logs are stored in PostgreSQL database as a source of truth.

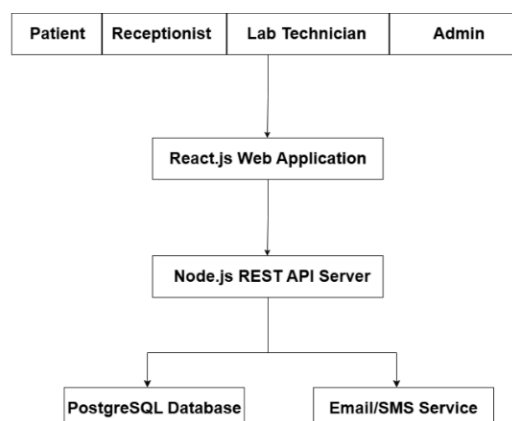
Thus, the suggested Laboratory Information System greatly increases laboratory performance by saving time spent on manual filling of paper reports, minimizing risks of errors in transcriptions, allowing standardization of reports, facilitating their generation, providing easy access to the data, allowing to make appointments online and receive notifications. In addition, the solution is efficient, scalable and inexpensive and suitable for labs working with semi-automatic equipment.

1.2 Illustrations Supporting the Narrative

In this section, you will find the diagrams that will be used for representing the proposed Laboratory Information System (LIS). Diagrams represent a visual description of the system structure, modules, user interface, and process. They show how the interaction between the users and the system occurs and how information is exchanged between the system elements.

1.2.1 Overall System Architecture

Figure 1: Overall Architecture



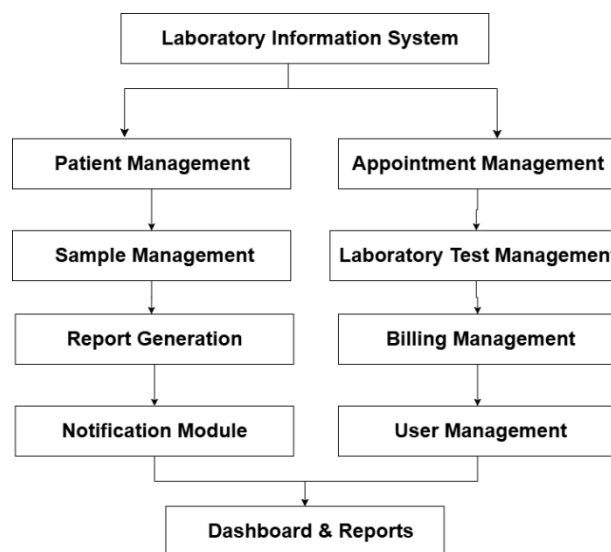
Overall Architecture of the Proposed Laboratory Information System

The system architecture diagram clearly shows the three-tier design used in the Laboratory Information System. The presentation tier includes the React.js front-end web application that offers an interactive interface for patients, receptionists, laboratory technicians, and administrators. The application tier is developed using the Node.js application that handles business logic, authentication, validations, and communication between the front-end and the database. The database tier includes the PostgreSQL database that manages patient data, appointments, laboratory test data, billing information, and users.

The use of this architecture has ensured that the user interface, application logic, and database are independent of each other.

1.2.2 Functional Modules of the System

Figure 2: Functional modules



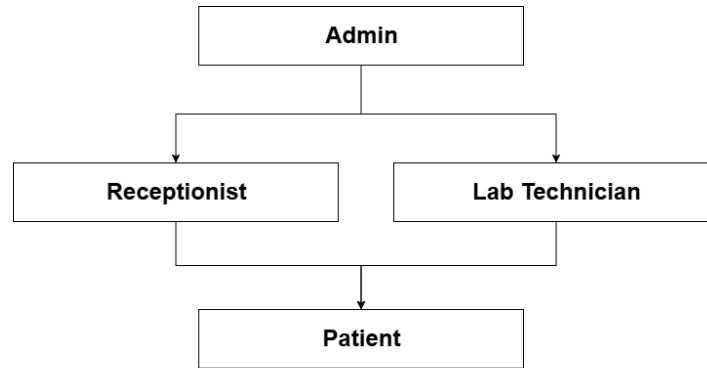
Main Functional Modules of the Proposed LIS

Functional Module Diagram explains the different modules of Laboratory Information System. The system consists of Appointment Management, Patient Management, Sample Management, Laboratory Test Management, Report Generation, Billing Management, Notification Management, User Management and Dashboard modules.

Each module is responsible for performing a particular task, however, all the modules share their data with each other via central database. This structure helps the system in being expandable and manageable.

1.2.3 User Roles and Responsibilities

Figure 3: User roles



User Roles

The user role diagram identifies the four primary users of the proposed Laboratory Information System and their responsibilities.

Table 1: User roles and responsibilities

User	Main Responsibilities
Administrator	Manages user accounts, system settings, laboratory services, reports, overall system administration
Receptionist	Registers patients, schedules appointments, manages patient information, records sample collection details, handles billing activities
Technician	Processes laboratory samples, manually enters test results obtained from semi-automatic equipment, verifies laboratory data, submits results for report generation
Patient	Books appointments online, views appointment status, receives notifications, accesses completed laboratory reports through the patient portal

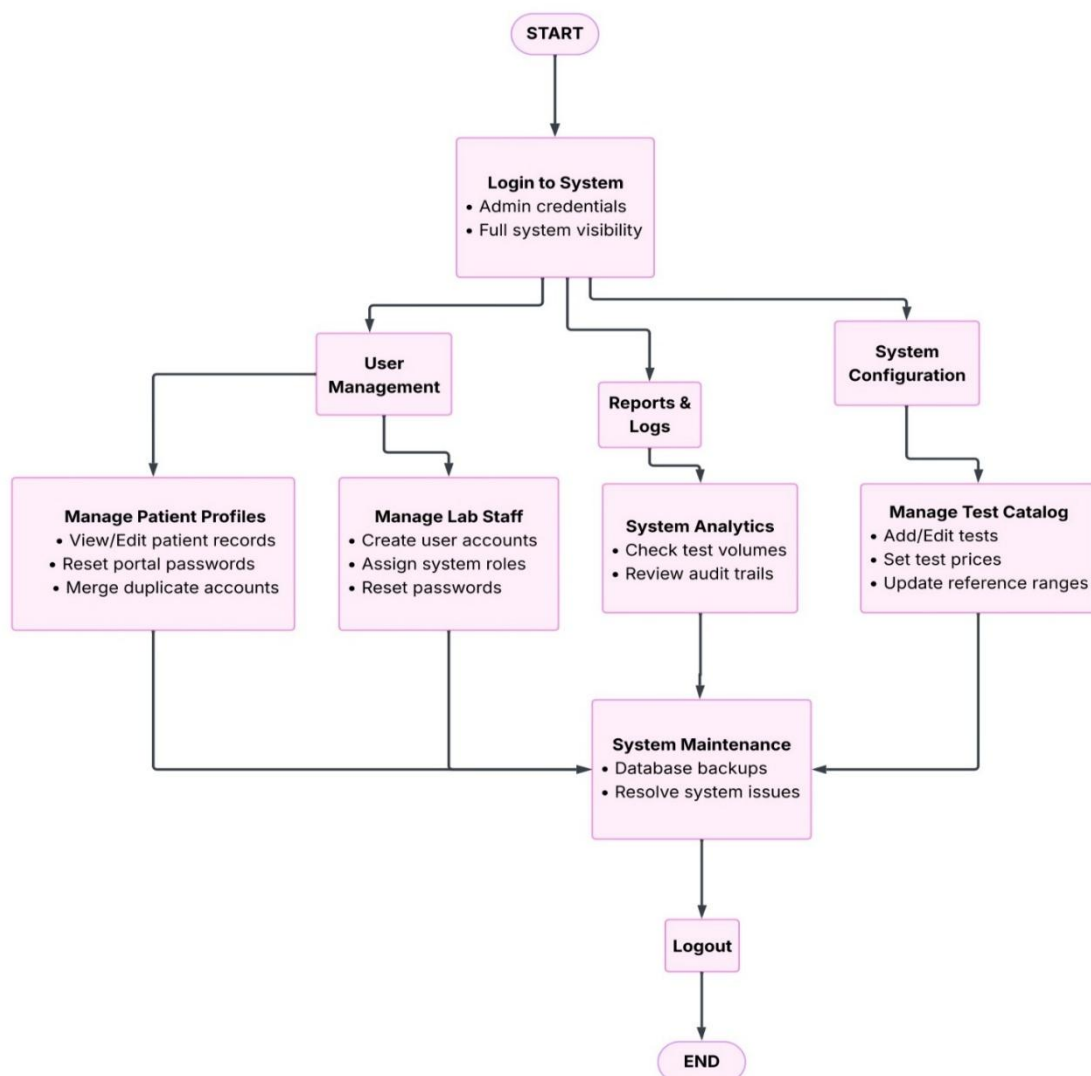
1.3 Process flows

The process flow diagrams will be used to demonstrate the order in which tasks have been done using the LIS system. These diagrams display how information flows into and through the system and how various users will use the system to perform their respective functions within the laboratories. They will aid in illustrating the business processes.

1.3.1 Administrator Workflow

Administrator Workflow is an example of tasks carried out by the system administrator. They involve user administration, laboratory tests administration, reports monitoring, system settings and ensuring system security. The administrator has all the privileges to administer the system resources for proper functioning of the laboratory.

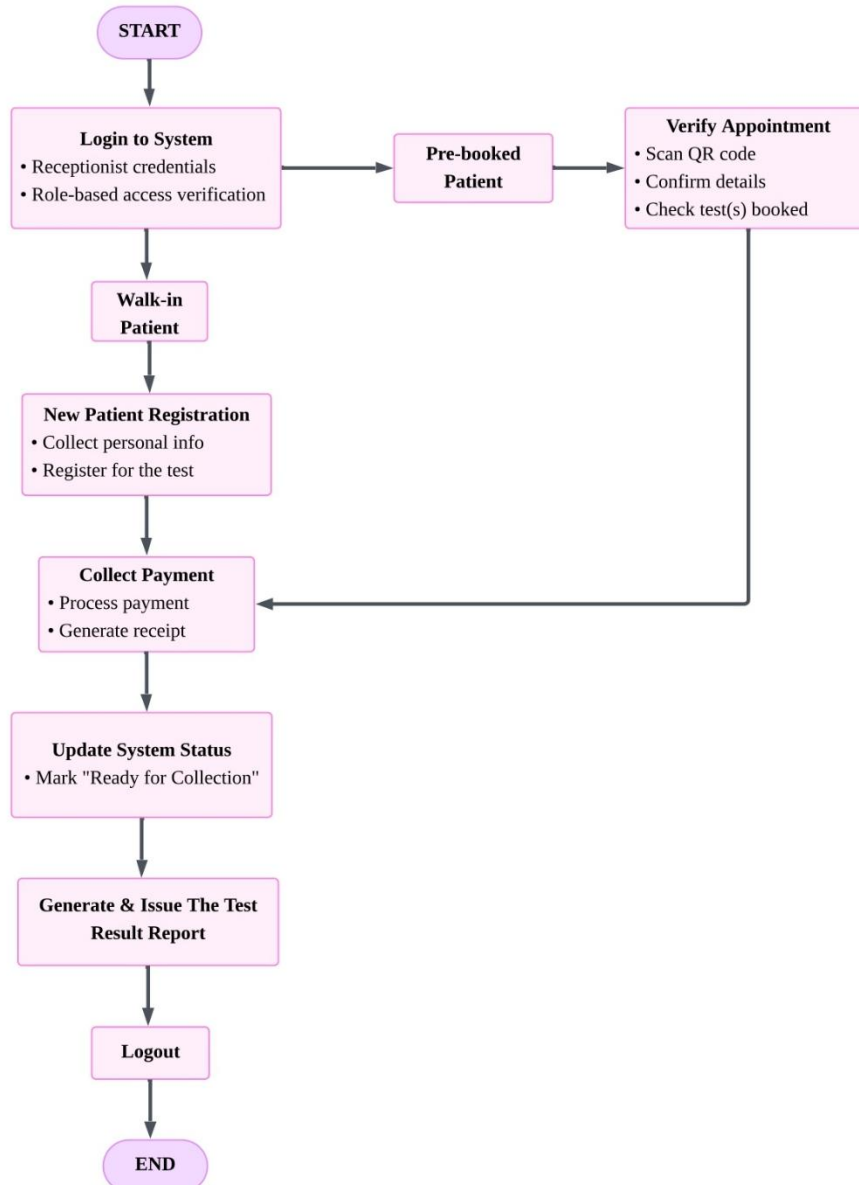
Figure 4: Admin workflow



1.3.2 Receptionist Workflow

The Patient Activities workflow is defined by how one goes about handling patients. This is because the receptionist performs duties such as registering the patient, scheduling appointments, recording samples collected, issuing invoices, and updating payment information.

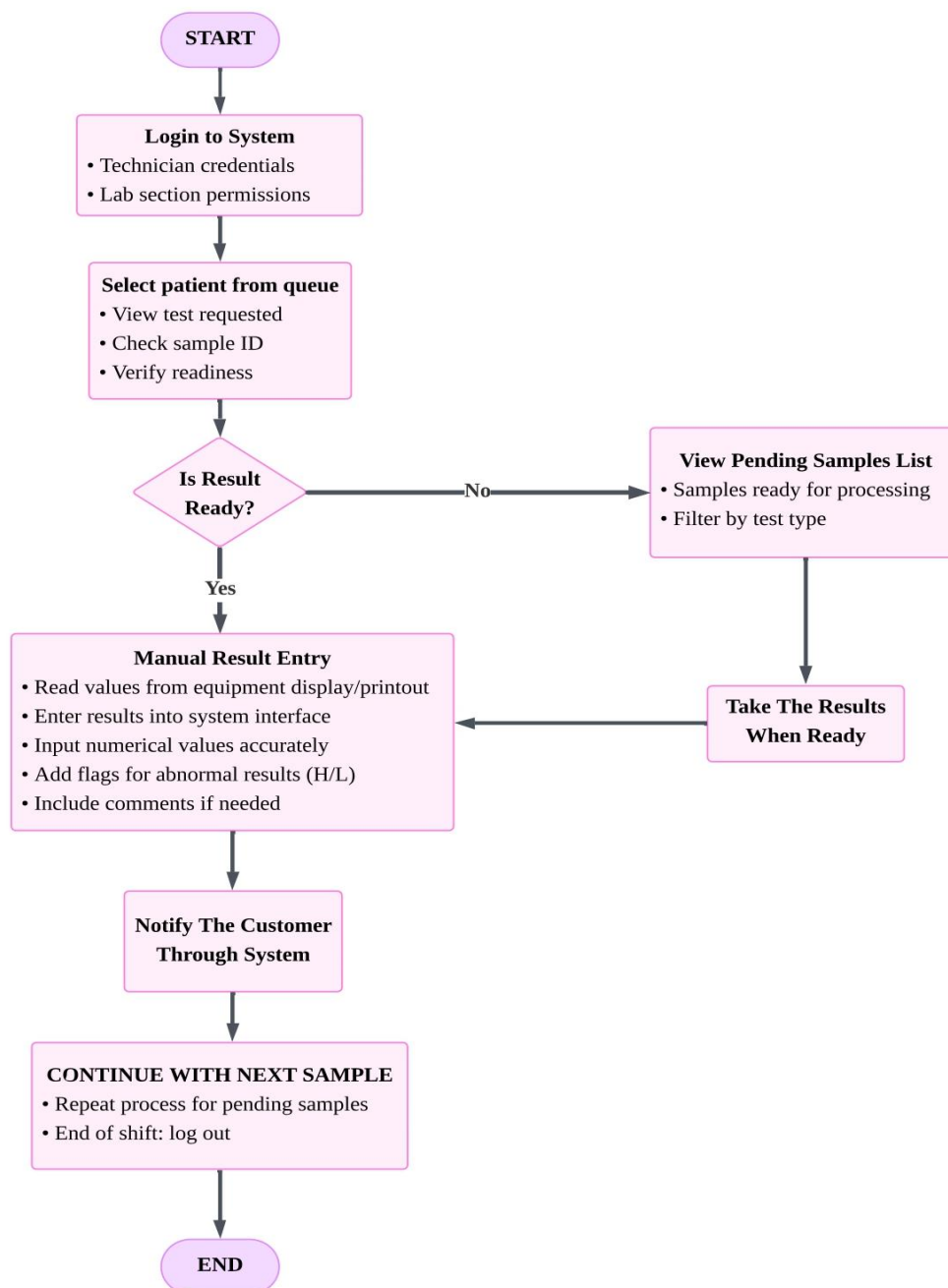
Figure 5: Receptionist workflow diagram



1.3.4 Laboratory Technician workflow

The Laboratory Technician workflow begins with receiving collected samples. After performing laboratory tests using semi-automatic equipment, the technician manually enters the test results into the system. Once the results are verified, they are submitted for report generation. This workflow reduces transcription errors and maintains accurate laboratory records.

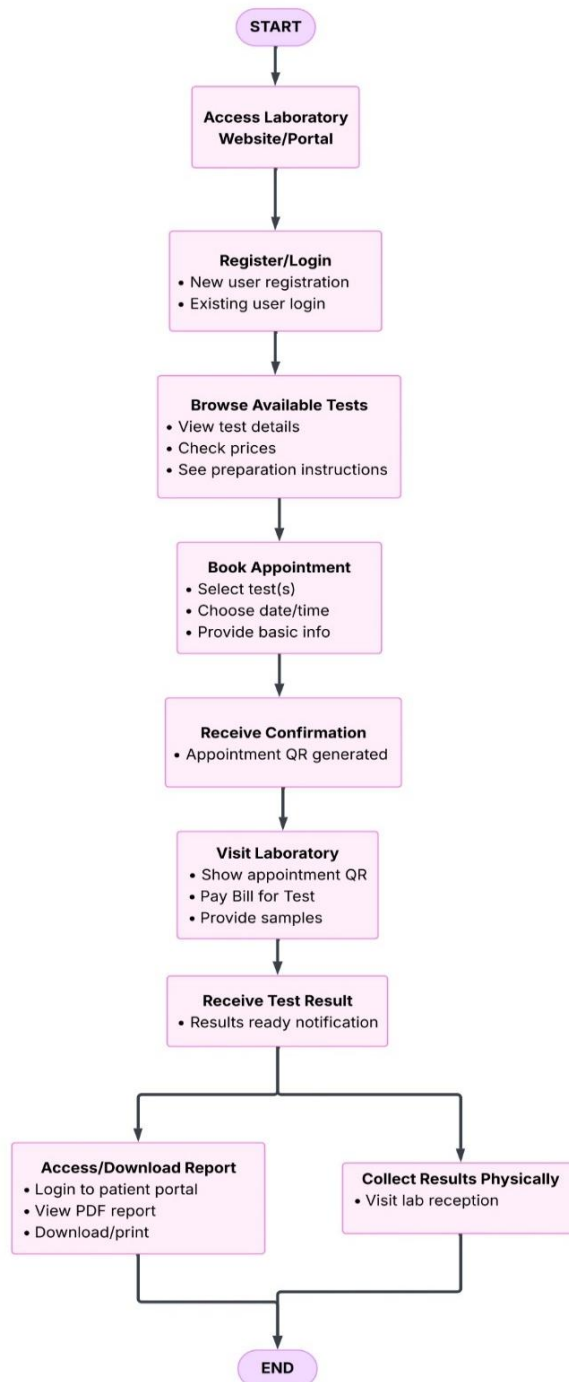
Figure 6: Lab technician workflow



1.3.5 Patient Workflow

The Patient workflow illustrates the services available to patients through the system. Patients can book appointments, view appointment details, receive SMS or email notifications, and access completed laboratory reports through the patient portal. This workflow improves patient convenience by reducing the need for physical visits to obtain laboratory reports.

Figure 7: Patient workflow



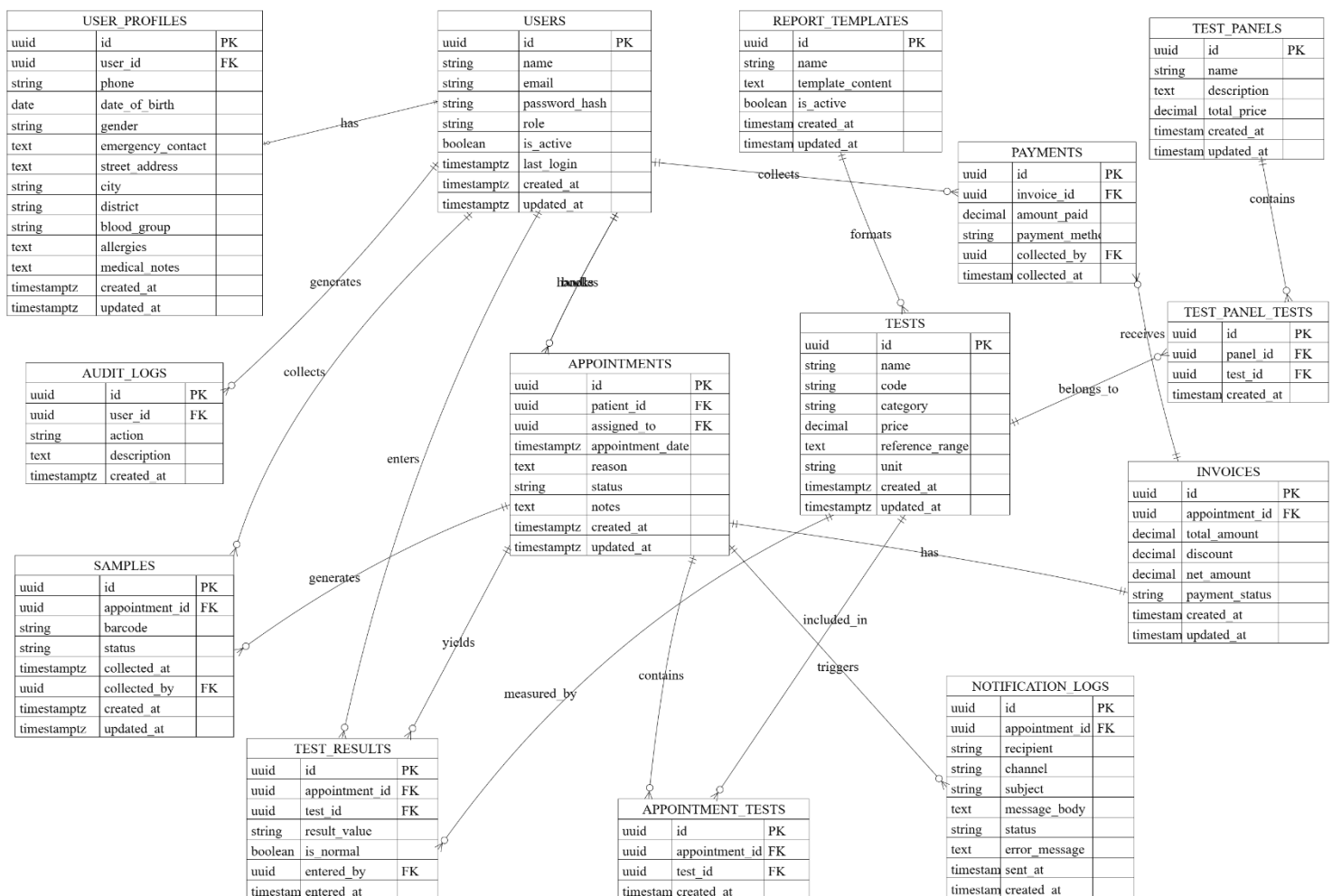
Chapter 2: Detailed design

2.1 Data Design

Data design is a critical component of the Laboratory Information System design process. It defines how data will be structured, stored, and managed within the system. This section presents the Entity-Relationship Diagram, the comprehensive Data Dictionary, and the Normalisation process that ensures data integrity and minimises redundancy. The data design is based on the requirements identified during the analysis phase and supports all functional modules of the system.

2.1.1 Entity-Relationship Diagram

Figure 8: ER diagram



The Entity-Relationship (ER) Diagram illustrates the logical structure of the database used by the Laboratory Information System. It identifies the key entities, their attributes, and the relationships between them. The diagram serves as a blueprint for the database schema implementation in PostgreSQL.

The Laboratory Information System database consists of ten primary entities: Users, Patients, Appointments, LaboratoryTests, Samples, TestResults, Reports, Billing, Notifications, and AuditLogs. Each entity is designed to store specific information relevant to laboratory operations, and the relationships between entities ensure data consistency and referential integrity.

The Users entity stores authentication and profile information for all system users, including administrators, receptionists, technicians, and patients. Each user record is linked to a role that determines their access privileges within the system. The Patients entity extends user information with medical-specific details such as date of birth, gender, blood type, and emergency contact information.

The Appointments entity records patient bookings for laboratory tests, including the preferred date and time, reason for the test, and a unique QR token generated for appointment verification at the reception. The LaboratoryTests entity maintains the catalogue of available diagnostic tests with their categories, reference ranges, measurement units, and pricing.

The Samples entity tracks specimen collection with unique barcodes, while the TestResults entity stores the actual test values entered by technicians. The Reports entity manages the generated PDF laboratory reports, and the Billing entity handles invoicing and payment records. The Notifications entity logs all SMS and email communications sent to patients, and the AuditLogs entity records all significant system activities for security and compliance purposes.

2.1.2 Data Dictionary

The data dictionary provides a detailed description of each data element stored in the Laboratory Information System database. It specifies field names, data types, constraints, and descriptions for every attribute in each entity. This comprehensive reference ensures consistency in data handling and serves as a guide for database implementation and maintenance.

Users Table

The Users table stores authentication credentials and basic profile information for all system users. It serves as the central identity management entity, supporting role-based access control.

Table 2: User table

Attribute	Data Type	Key	Constraints	Description
id	UUID	PK	Default gen_random_uuid()	Unique identifier for each user.
name	VARCHAR(255)	-	NOT NULL	Full name of the user.
email	VARCHAR(255)	-	NOT NULL, UNIQUE	User's email address used for login.
password_hash	VARCHAR(255)	-	NOT NULL	Encrypted user password.
role	VARCHAR(50)	-	NOT NULL, CHECK	User role (Admin, Receptionist, Technician, Patient).
is_active	BOOLEAN	-	Default TRUE	Indicates whether the account is active.
last_login	TIMESTAMPTZ	-	Nullable	Date and time of last login.
created_at	TIMESTAMPTZ	-	Default NOW()	Record creation timestamp.
updated_at	TIMESTAMPTZ	-	Default NOW()	Record update timestamp.

User Profile Table

Stores additional personal and medical information related to each user. It extends the user account with contact details, address, and health-related information.

Table 3: User profile table

Attribute	Data Type	Key	Constraints	Description
id	UUID	PK	Default gen_random_uuid()	Unique profile identifier.
user_id	UUID	FK	UNIQUE, REFERENCES USERS(id)	Associated user account.
phone	VARCHAR(50)	-	Nullable	Contact number.
date_of_birth	DATE	-	Nullable	Date of birth.
gender	TEXT	-	CHECK	Gender of the user.

emergency_contact	TEXT	-	Nullable	Emergency contact details.
street_address	TEXT	-	Nullable	Street address.
city	TEXT	-	Nullable	City.
district	TEXT	-	Nullable	District.
blood_group	TEXT	-	Nullable	Blood group.
allergies	TEXT	-	Nullable	Allergy information.
medical_notes	TEXT	-	Nullable	Additional medical information.
created_at	TIMESTAMPTZ	-	Default NOW()	Creation timestamp.
updated_at	TIMESTAMPTZ	-	Default NOW()	Last updated timestamp.

Appointments Table

The Appointments table records all patient appointments for laboratory tests, tracking the appointment lifecycle from booking through completion.

Table 4: Appointment table

Attribute	Data Type	Key	Constraints	Description
id	UUID	PK	Default gen_random_uuid()	Appointment identifier.
patient_id	UUID	FK	REFERENCES USERS(id)	Patient who booked the appointment.
assigned_to	UUID	FK	REFERENCES USERS(id)	Assigned laboratory staff.
appointment_date	TIMESTAMPTZ	-	NOT NULL	Appointment date and time.
reason	TEXT	-	Nullable	Reason for appointment.
status	VARCHAR(50)	-	CHECK	Appointment status.
notes	TEXT	-	Nullable	Additional notes.
created_at	TIMESTAMPTZ	-	Default NOW()	Creation timestamp.

updated_at	TIMESTAMPTZ	-	Default NOW()	Last updated timestamp.
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Tests Table

The LaboratoryTests table maintains the catalogue of all diagnostic tests offered by the laboratory, including medical categories, pricing, and reference ranges.

Table 5: Tests table

Attribute	Data Type	Key	Constraints	Description
id	UUID	PK	Default gen_random_uuid()	Test identifier.
name	VARCHAR(255)	-	UNIQUE	Test name.
code	VARCHAR(50)	-	UNIQUE	Test code.
category	VARCHAR(100)	-	NOT NULL	Test category.
price	DECIMAL(10,2)	-	NOT NULL	Test price.
reference_range	TEXT	-	NOT NULL	Normal reference range.
unit	VARCHAR(50)	-	NOT NULL	Measurement unit.
created_at	TIMESTAMPTZ	-	Default NOW()	Creation timestamp.
updated_at	TIMESTAMPTZ	-	Default NOW()	Last updated timestamp.

Samples Table

The Samples table tracks all specimen collections with unique barcodes for sample identification and status tracking throughout the testing process.

Table 6: Samples table

Attribute	Data Type	Key	Constraints	Description
id	UUID	PK	Default gen_random_uuid()	Sample identifier.
appointment_id	UUID	FK	REFERENCES APPOINTMENTS(id)	Related appointment.
barcode	VARCHAR(100)	-	UNIQUE	Sample barcode.

status	VARCHAR(50)	-	CHECK	Sample processing status.
collected_at	TIMESTAMPTZ	-	Nullable	Collection date and time.
collected_by	UUID	FK	REFERENCES USERS(id)	Technician collecting sample.
created_at	TIMESTAMPTZ	-	Default NOW()	Creation timestamp.
updated_at	TIMESTAMPTZ	-	Default NOW()	Last updated timestamp.

Test Results Table

The Test results table stores the actual laboratory test values entered by technicians, including automatic flagging for abnormal results.

Table 7: Test results table

Attribute	Data Type	Key	Constraints	Description
id	UUID	PK	Default gen_random_uuid()	Result identifier.
appointment_id	UUID	FK	REFERENCES APPOINTMENTS(id)	Appointment identifier.
test_id	UUID	FK	REFERENCES TESTS(id)	Test identifier.
result_value	VARCHAR(255)	-	NOT NULL	Laboratory result value.
is_normal	BOOLEAN	-	Default TRUE	Indicates whether the result is within the normal range.
entered_by	UUID	FK	REFERENCES USERS(id)	Technician entering the result.
entered_at	TIMESTAMPTZ	-	Default NOW()	Result entry timestamp.

Reports Table

The Reports table manages the automatically generated PDF laboratory reports, linking them to appointments and tracking their status.

Table 8: Report table

Field Name	Data Type	Key	Constraints	Description
report_id	UUID	PK	Default gen_random_uuid()	Unique identifier for each report.
appointment_id	UUID	FK	REFERENCES USERS(id)	References the associated appointment.
generated_at	TIMESTAMP	-	DEFAULT NOW()	Date and time when the report was generated.
pdf_path	VARCHAR(255)	-	NOT NULL	File path or URL of the generated PDF report.
status	ENUM	-	NOT NULL	Report status (GENERATED, DELIVERED, ARCHIVED).
generated_by	UUID	FK	UNIQUE	References the user (technician or administrator) who generated the report.

Billing Table

The Billing table handles all financial transactions, recording invoices and payment details for each appointment.

Table 9: Billing table

Field Name	Data Type	Key	Constraints	Description
invoice_id	UUID	PK	Default gen_random_uuid()	Unique identifier for each invoice
appointment_id	UUID	FK	REFERENCES USERS(id)	Reference to Appointments table
amount	DECIMAL(10,2)	-	NOT NULL	Total invoice amount in LKR
payment_status	ENUM	-	NOT NULL	PENDING, PAID, REFUNDED
payment_method	ENUM	-	NULLABLE	CASH, CARD, ONLINE
paid_at	TIMESTAMP	-	NULLABLE	Payment completion timestamp

Notifications Table

The Notifications table logs all automated communications sent to patients, including SMS and email notifications for report availability.

Table 10: Notification table

Field Name	Data Type	Key	Constraints	Description
notification_id	UUID	PK	Default gen_random_uuid()	Unique identifier for each notification
appointment_id	UUID	FK	REFERENCES USERS(id)	Reference to Appointments table
type	ENUM	-	NOT NULL	SMS, EMAIL, IN_APP
recipient	VARCHAR(100)	-	NOT NULL	Phone number or email address
sent_at	TIMESTAMP	-	DEFAULT NOW()	Notification dispatch timestamp
status	ENUM	-	NOT NULL	SENT, FAILED, PENDING

Audit Logs Table

The AuditLogs table maintains a comprehensive record of all significant system activities for security auditing and compliance purposes.

Table 11: Audit logs table

Attribute	Data Type	Key	Constraints	Description
id	UUID	PK	Default gen_random_uuid()	Audit log identifier.
user_id	UUID	FK	REFERENCES USERS(id)	User performing the action.
action	TEXT	-	NOT NULL	Performed action.
description	TEXT	-	NOT NULL	Action description.
created_at	TIMESTAMPTZ	-	Default NOW()	Log creation timestamp.

2.1.3 Normalisation

Normalisation is the process of organising database columns and tables to minimise data redundancy and improve data integrity. The Laboratory Information System database has been normalised to Third Normal Form (3NF) to ensure optimal structure and performance.

First Normal Form (1NF)

All tables in the database satisfy First Normal Form. Each column contains only atomic (indivisible) values, and each column contains values of a single type. There are no repeating groups or arrays within any table. For example, the Users table stores each attribute (first_name, last_name, email) in separate columns rather than combining them. The Patients table stores allergies as a single text field rather than multiple columns, since allergies are optional and variable in number.

Second Normal Form (2NF)

All tables satisfy Second Normal Form. Every non-key attribute is fully functionally dependent on the entire primary key. This is particularly important for tables with composite keys. In the TestResults table, the observed_value, flag, and entered_at attributes depend on the result_id primary key and not on any partial key. All multi-field tables have been examined to ensure no partial dependencies exist.

Third Normal Form (3NF)

All tables satisfy Third Normal Form. There are no transitive dependencies where a non-key attribute depends on another non-key attribute. For example, test pricing information is stored directly in the LaboratoryTests table rather than being derived from the test category. Patient demographic information is stored in the Patients table rather than being duplicated in the Appointments table.

The following normalisation summary demonstrates how potential redundancies were eliminated:

Table 12: Normalisation summary

Original Structure	Normalisation Applied	Resulting Structure	Benefit
Patient details stored in Appointments	Extracted to Patients table with FK	Separate Patients and Appointments tables	Patient details stored once, updated centrally
Test info stored with each result	Extracted to LaboratoryTests table	Separate LaboratoryTests and TestResults tables	Test catalogue maintained independently

User credentials in role-specific tables	Centralised in Users table with role field	Single Users table, role-specific tables reference it	Unified authentication, reduced redundancy
Billing info in Appointments table	Extracted to Billing table	Separate Billing table with appointment FK	Independent payment tracking and reporting
Report data duplicated from results	Reports table references appointment	Reports stores only metadata and PDF path	Eliminates data duplication, stores only generated output

The normalisation process ensures that the database maintains data integrity while minimising storage redundancy. The resulting schema supports efficient query performance and simplifies database maintenance operations. Where performance optimisation is required for reporting queries, appropriate database indexes have been defined on foreign key columns and frequently queried fields.

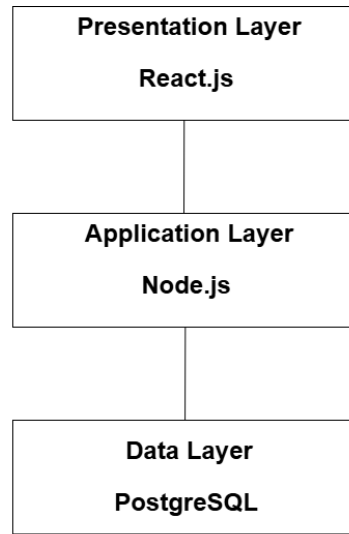
2.2 Architectural Design

An architectural design provides the structure of the software by establishing major components of the software system, their functions, and interactions among them. The architectural design offers guidelines for developing the software by grouping software into various components. An effective architectural design enhances maintainability, scalability, security, and performance of the software, and makes future developments easy.

The designed Laboratory Information System (LIS) utilizes a three-layer architecture which includes Presentation Layer, Application Layer, and Data Layer. The three-layer architecture ensures that the user interface, business logic, and data management are separated into various layers whereby the layers accomplish distinct responsibilities without being dependent on how the other layers have been implemented. The interactions between layers occur through secure RESTful APIs.

2.2.1 Three-Tier Architecture

Figure 9: Three- tire architecture



Three-Tier Technology Architecture

The proposed Laboratory Information System follows a three-tier architecture to ensure clear separation of responsibilities and improve system maintainability. Each layer performs a dedicated function while communicating only with its adjacent layer.

1. Presentation Layer

The Presentation Layer provides the graphical user interface through which users interact with the Laboratory Information System. It is developed using React.js and Tailwind CSS, providing a responsive and user-friendly web interface accessible from desktop computers, tablets, and mobile devices.

Users:

- i. Administrator
- ii. Receptionist
- iii. Laboratory Technician
- iv. Patient

Responsibilities:

- | | |
|---|--|
| i. Display user interfaces. | v. Send user requests to the Application Layer. |
| ii. Capture user inputs. | vi. Display responses received from the backend. |
| iii. Validate basic form data. | |
| iv. Display laboratory reports and notifications. | |

Main Interfaces:

- | | |
|---------------------------|-------------------------|
| i. Login | vi. Result Entry |
| ii. Dashboard | vii. Billing |
| iii. Patient Registration | viii. Report Generation |
| iv. Appointment Booking | ix. Patient Portal |
| v. Sample Registration | |

The Presentation Layer does not communicate directly with the database. Instead, all requests are forwarded securely to the Application Layer.

2. Application Layer

The Application Layer contains the core business logic of the Laboratory Information System. It is implemented using Node.js with Express.js and acts as the intermediary between the Presentation Layer and the Data Layer.

This layer processes user requests, applies laboratory business rules, validates data, performs authentication, and coordinates communication between different system modules.

Main Components:

- | | |
|-----------------------------------|------------------------------|
| i. Authentication Module | v. Laboratory Test Module |
| ii. Appointment Management Module | vi. Report Generation Module |
| iii. Patient Management Module | vii. Billing Module |
| iv. Sample Management Module | viii. Notification Module |
| | ix. User Management Module |

Responsibilities:

- | | |
|--|--|
| i. Process user requests. | vi. Send SMS and email notifications. |
| ii. Authenticate system users using JWT. | vii. Coordinate communication between modules. |
| iii. Validate patient and laboratory data. | viii. Retrieve and update data through the Data Layer. |
| iv. Generate laboratory reports. | |
| v. Calculate billing information. | |

This layer represents the intelligence of the system by enforcing laboratory workflows and ensuring that all operations comply with the required business rules.

3. Data Layer

The Data Layer manages all persistent data used by the Laboratory Information System. The project uses PostgreSQL as the relational database management system.

Responsibilities:

- | | |
|---|----------------------------------|
| i. Store patient records. | v. Store billing records. |
| ii. Store appointments. | vi. Store user accounts. |
| iii. Store laboratory test information. | vii. Store notification history. |
| iv. Store laboratory reports. | viii. Maintain audit logs. |

The Data Layer responds to queries received from the Application Layer while ensuring data consistency, integrity, and security. Direct access to the database from the user interface is not permitted.

Information Flow Between Layers

Communication among the layers takes place in sequential order during all operations performed on the system. Whenever a user makes a request via the Presentation Layer, the request is sent to the Application Layer through RESTful API calls. The Application Layer verifies the request, does the needful logic operations and interacts with the PostgreSQL database through SQL queries. Finally, the output is sent back to the Presentation Layer for presentation to the user.

Through such a communication process among the layers, loose coupling can be achieved.

Benefits of Three-Tier Architecture

- i. Separation of user interface, business logic, and database operations.
- ii. Improved maintainability through modular development.
- iii. Enhanced system security by preventing direct database access.
- iv. Easier scalability for future system expansion.
- v. Simplified testing and debugging.
- vi. Better code reusability.
- vii. Improved performance and reliability.

2.2.2 System Modules

The proposed Laboratory Information System is divided into several independent modules. Each module performs a specific function while exchanging information with other modules through the Application Layer.

Table 13: System modules

Module	Purpose	Responsibilities
Authentication Module	Provides secure authentication and authorization for all users.	i. User login ii. Password verification iii. JWT authentication iv. Role-based access control v. Session management
Appointment Management Module	Manages laboratory appointment scheduling.	i. Book appointments ii. View appointments iii. Cancel appointments
Patient Management Module	Maintains patient information.	i. Register patients ii. Update patient details iii. Search patient records iv. Maintain patient history
Sample Management Module	Handles laboratory sample registration and tracking.	i. Generate sample barcode ii. Register collected samples iii. Track sample status iv. Link samples with patients
Laboratory Test Module	Records laboratory test information.	i. Assign laboratory tests ii. Enter test results iii. Validate laboratory results iv. Store completed tests
Report Generation Module	Automatically generates standardized laboratory reports.	i. Retrieve laboratory results ii. Generate PDF reports iii. Store reports iv. Retrieve historical reports
Billing Module	Manages laboratory payments.	i. Calculate laboratory charges ii. Generate invoices iii. Record payments iv. Generate receipts
Notification Module	Keeps patients informed about laboratory services.	i. Send SMS notifications ii. Send email notifications iii. Notify report availability iv. Record notification history
User Management Module	Allows administrators to manage system users.	i. Add users ii. Update users iii. Delete users iv. Assign user roles

2.2.3 Module Interaction

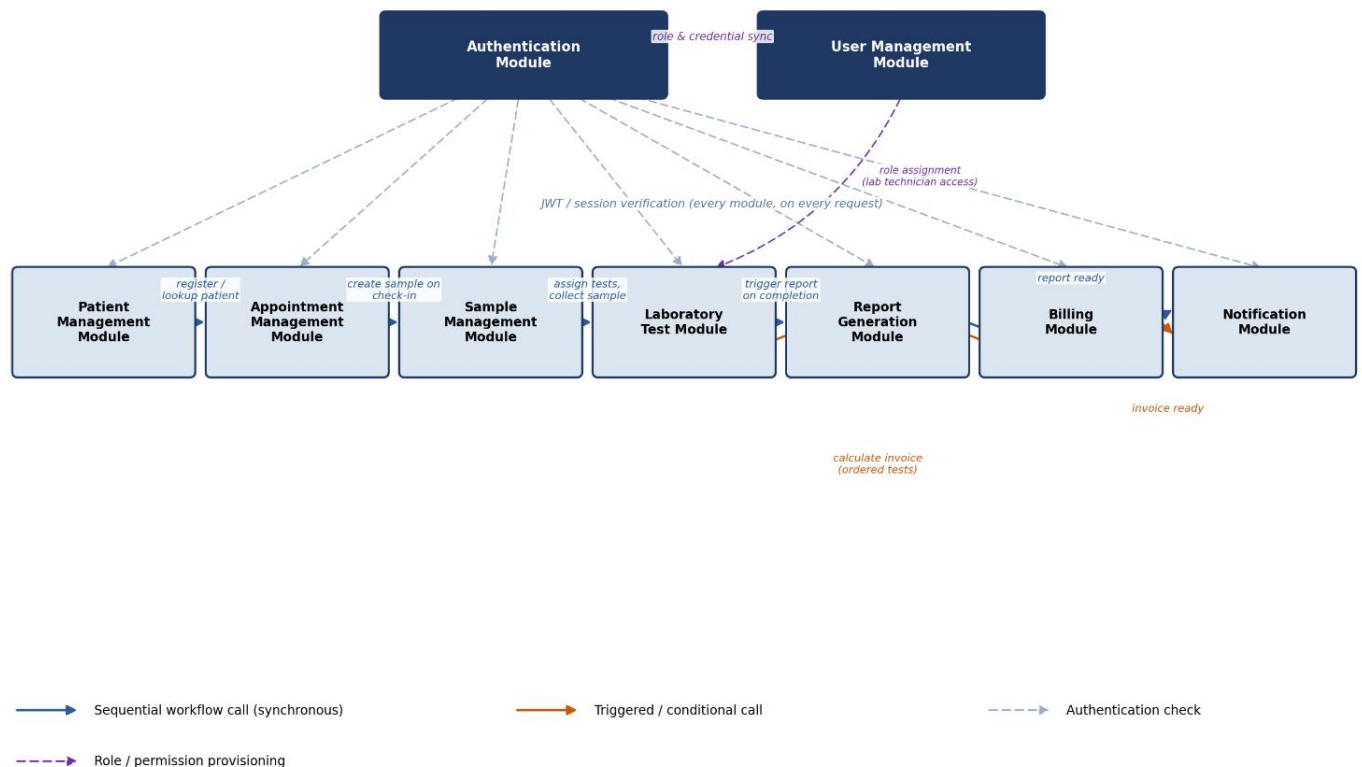
The modules in the Laboratory Information System use the Application Layer to perform tasks related to the laboratory work.

The Appointment Management Module creates appointment records, which are later used by the Patient Management Module to register patients. After patient registration is done, the Sample Management Module logs the collected sample and gives it a barcode number. Laboratory Test Module accesses the collected sample information and logs the laboratory test results provided by the laboratory technician.

After a successful process of result logging, the Report Generation Module automatically creates a laboratory report and saves it into the database. The Billing Module creates laboratory fees and invoices depending on chosen laboratory tests. Lastly, the Notification Module notifies patients about their report availability.

Unified communication between all modules avoids redundant data and helps to keep consistency.

Figure 10: Module interaction diagram



2.2.4 Use Case Diagram

The use case diagram illustrates the interactions between system users and the Laboratory Information System.

Figure 11: Usecase diagram

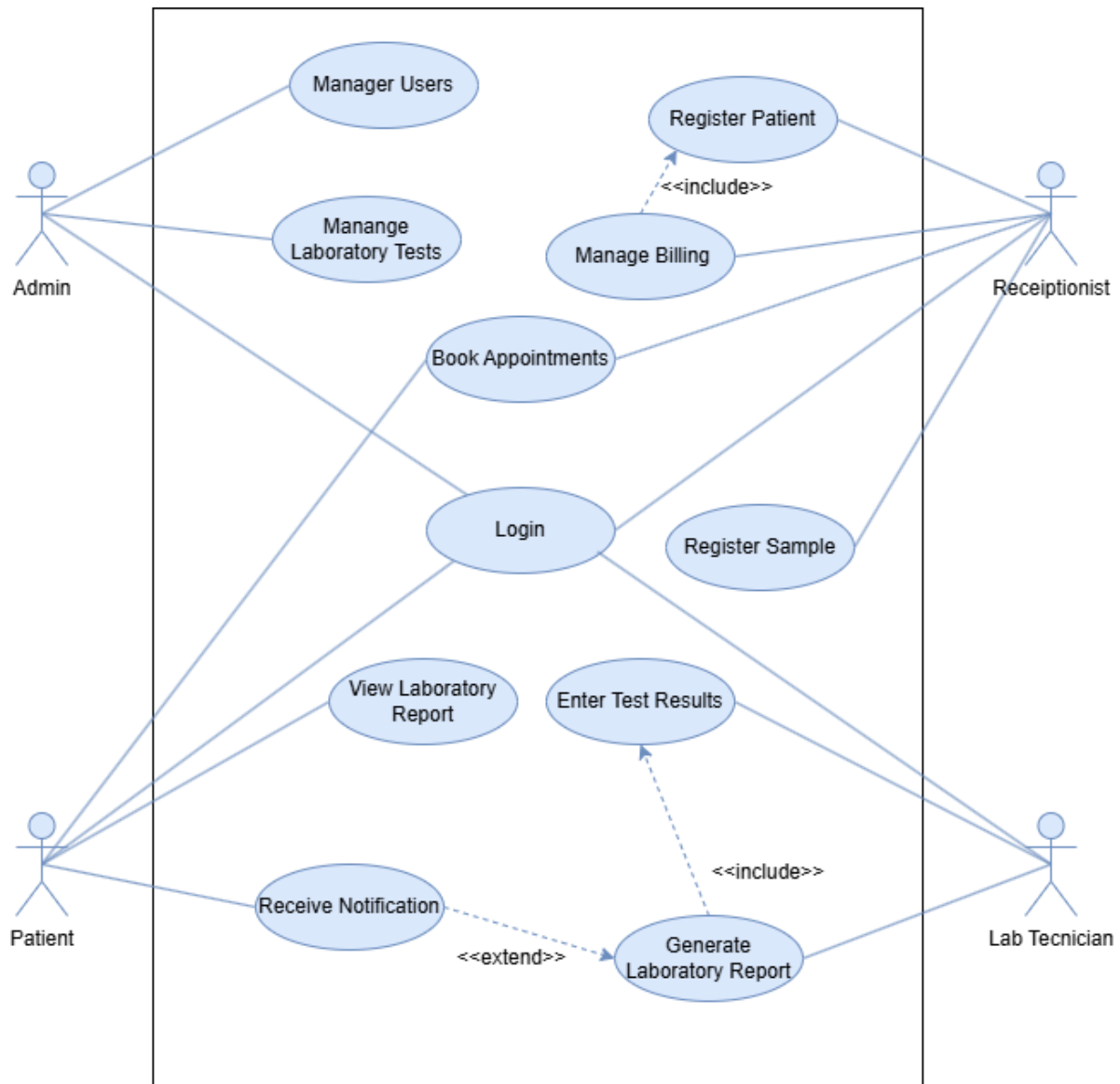


Table 14: Description about usecase

Use Case	Actor(s)	Description
Login	Administrator, Receptionist, Laboratory Technician, Patient	Authenticates users and grants access to the system according to their assigned roles.
Manage Users	Administrator	Allows administrators to create, update, deactivate, and manage user accounts and assign system roles.
Manage Laboratory Tests	Administrator	Allows administrators to add, update, or remove laboratory test definitions and pricing.
Register Patient	Receptionist	Registers new patients, updates patient information, and retrieves existing patient records.
Book Appointment	Patient, Receptionist	Allows appointments to be scheduled for laboratory tests by selecting available dates and times.
Register Sample	Receptionist	Records collected samples and assigns unique barcode identifiers.
Enter Test Results	Laboratory Technician	Records laboratory test results obtained from semi-automatic laboratory equipment.
Generate Laboratory Report	Laboratory Technician	Generates standardized laboratory reports based on validated test results.
Manage Billing	Receptionist	Calculates laboratory charges, generates invoices, and records payments.
View Laboratory Report	Patient	Allows patients to securely access completed laboratory reports through the patient portal.
Receive Notification	Patient	Receives SMS or email notifications informing them laboratory reports are available.

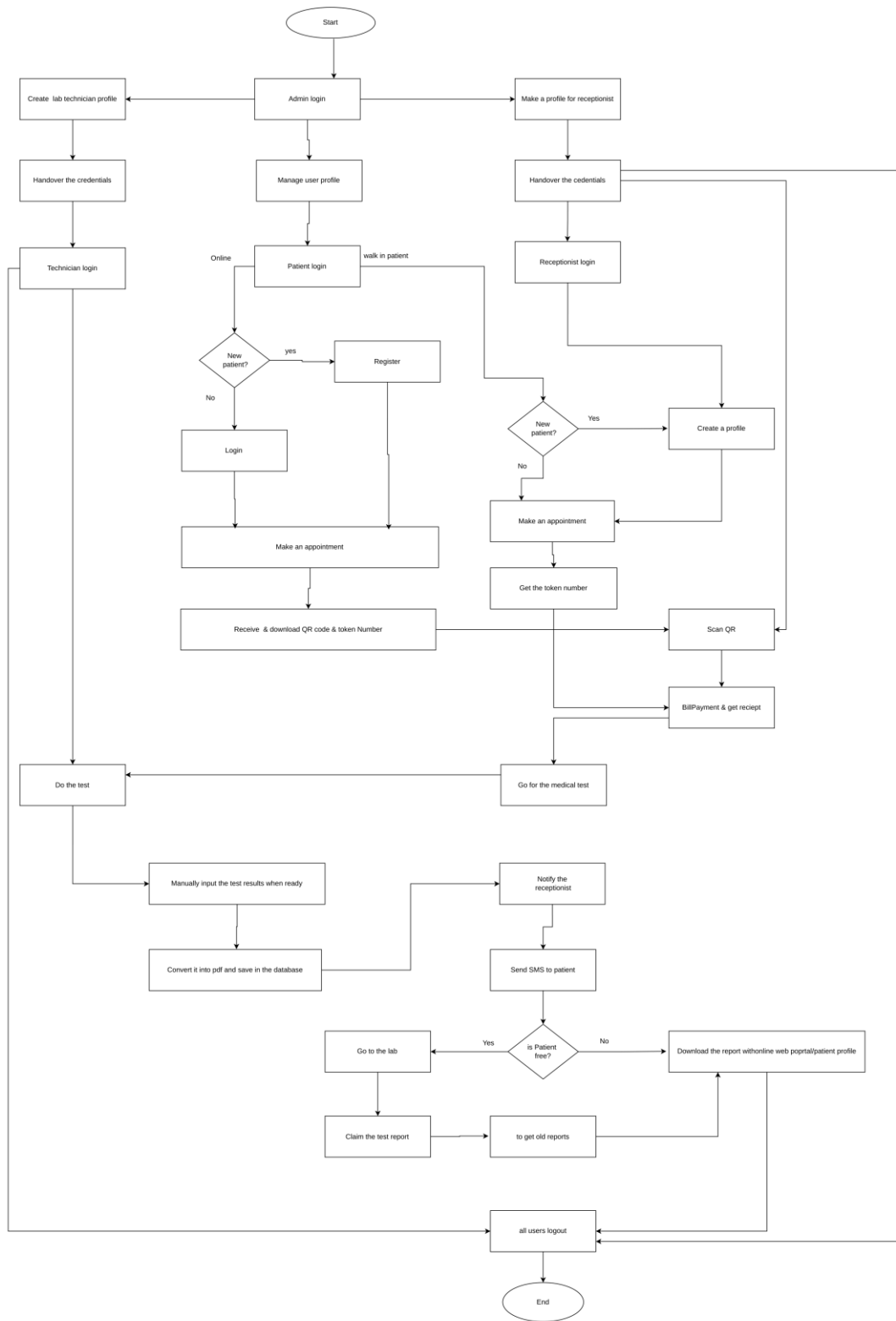
2.3 Procedural Design

Procedural Design divides the software system into small sub-routines and functions, representing the operation and logic of each module. It outlines procedures, logic, decisions, input/output, and flow control of the system in an organized manner and helps implement the logic of each module in a precise way. In this project, the procedural design of the four main modules of the Medical Laboratory Information System (LIS) is provided, which include Patient Registration & Appointment Booking, Receptionist Verification & Billing, Lab Technician Result Entry, and Report Generation & Notification. The logic of each module is shown in flow chart and activity diagram form.

2.3.1 Overall System Workflow

The entire workflow process of the Laboratory Information System is the total process that goes into running the Laboratory Information System. This process starts from booking an appointment by the patient to delivering the final laboratory report to the patient. During this process, different activities are carried out by the Receptionist, Laboratory Technician, and the Administrator while the system keeps the record and does everything automatically.

Figure 12: Overall workflow



User logs into the system

System identifies user role (Patient / Receptionist / Lab Technician / Admin)

IF ROLE = Patient

Show Patient Portal

Options:

- > Register / Login
- > Book Appointment
- > View / Download Report
- > Logout

Perform selected action

Return to portal or logout

IF ROLE = Receptionist

Show Receptionist Dashboard

Options:

- > Register Walk-in Patient
- > Verify Pre-booked Appointment (scan QR)
- > Collect Payment
- > Update Sample Status

Perform selected action

Return to dashboard or logout

IF ROLE = Lab Technician

Show Technician Dashboard

Options:

- > View Pending Sample Queue
- > Enter Test Result
- > Flag Abnormal Values

System generates reports and notifies patient/receptionist

Return to dashboard or logout

IF ROLE = Admin

Show Admin Dashboard

Options:

- > Create Receptionist / Technician Profile
- > Manage User Accounts

Perform selected action

Return to dashboard or logout

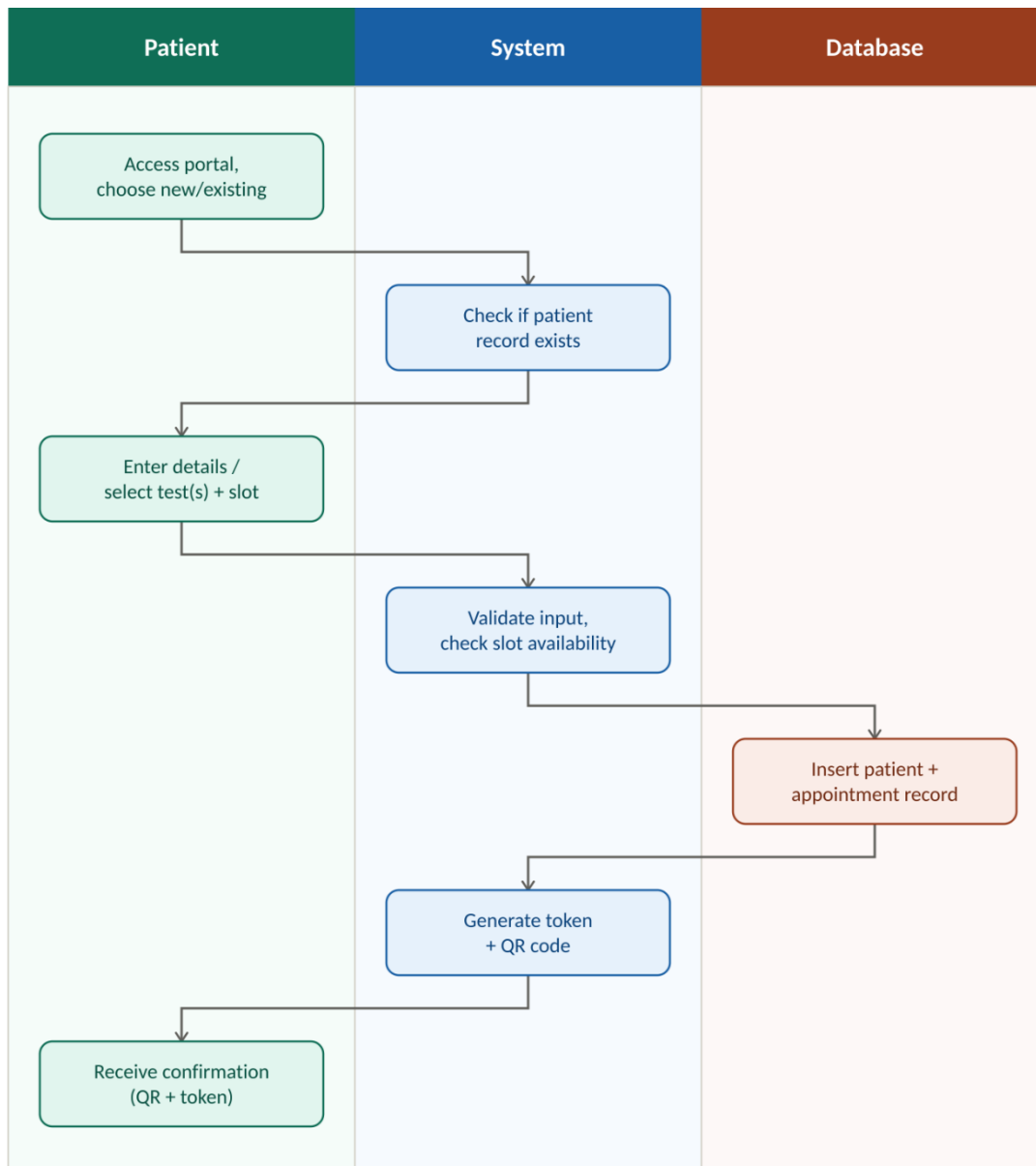
END / LOGOUT

2.3.2 Activity Diagram

The following activity diagrams present the swim lane-level logic of the four key processes in the system, showing the responsibilities of each actor alongside the System and Database.

Patient Registration & Appointment Booking

Figure 13: Activity diagram - Patient registration & appointment booking

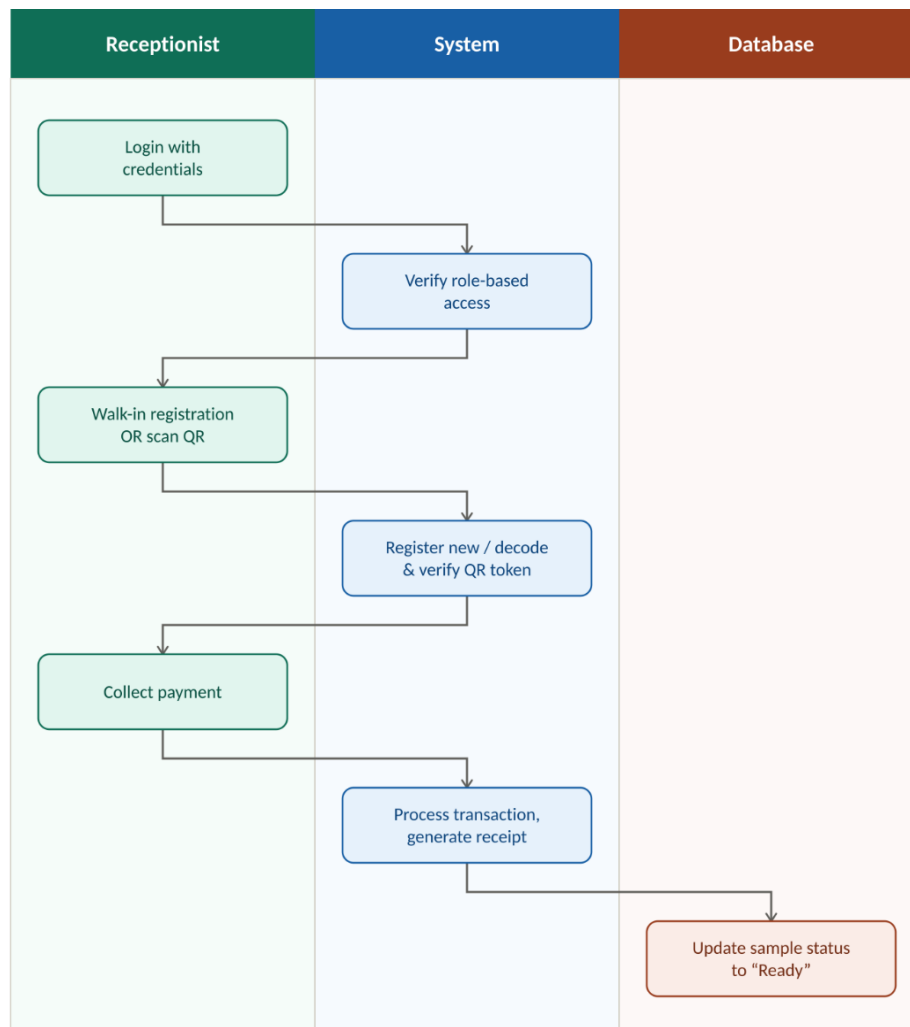


Key steps explained:

1. **Login/Registration** - The process begins when the patient accesses the portal and the system checks whether they are a new or existing user.
2. **New Patient Flow** - If new, the patient enters personal details; the system validates the input (e.g. required fields, valid contact number) before creating the record.
3. **Book Appointment** - The patient selects the test(s) and a preferred date/time slot. The system checks slot availability before proceeding, rejecting the booking and prompting an alternative slot if unavailable.
4. **Confirmation** - On success, the system generates a unique appointment token and QR code, stores the appointment record, and returns the confirmation to the patient.
5. **End of process** - The patient may later use this QR code at the reception to verify their appointment

Receptionist Verification & Billing

Figure 14: Activity diagram - receptionist verification & billing

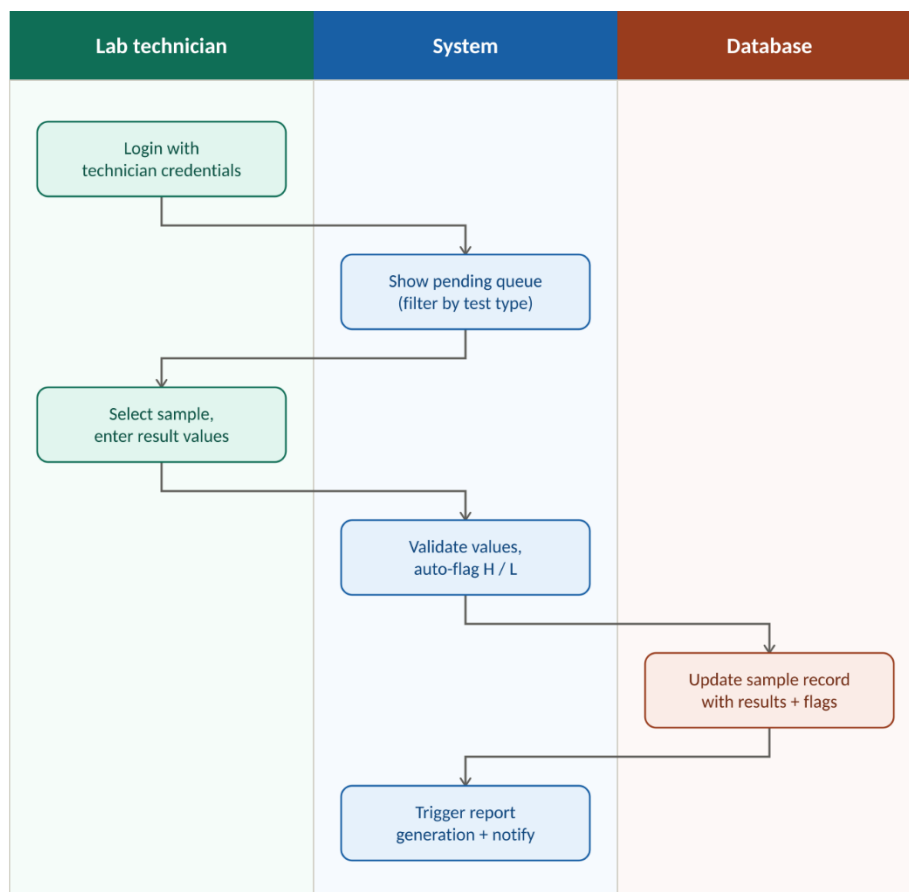


Key steps explained:

1. **Login** - The receptionist logs in; the system verifies role-based access before granting dashboard access.
2. **Decision: walk-in vs pre-booked** - This is the key branch point taken directly from the Receptionist Workflow: walk-in patients are registered fresh, while pre-booked patients are verified via QR code scan.
3. **Verify Appointment** - For pre-booked patients, the system decodes the QR token and retrieves the matching appointment; an invalid or expired code is rejected with an error message rather than allowing the process to continue.
4. **Collect Payment** - Payment is processed and a receipt generated only after a successful transaction; the sample status is not updated until payment is confirmed, which is a deliberate safeguard against unpaid samples entering the lab queue.
5. **Update Status** - Once payment is confirmed, the system marks the sample "Ready for Collection", which pushes it onto the Lab Technician's pending queue.

Lab Technician Result Entry

Figure 15: Activity diagram - lab technician result entry

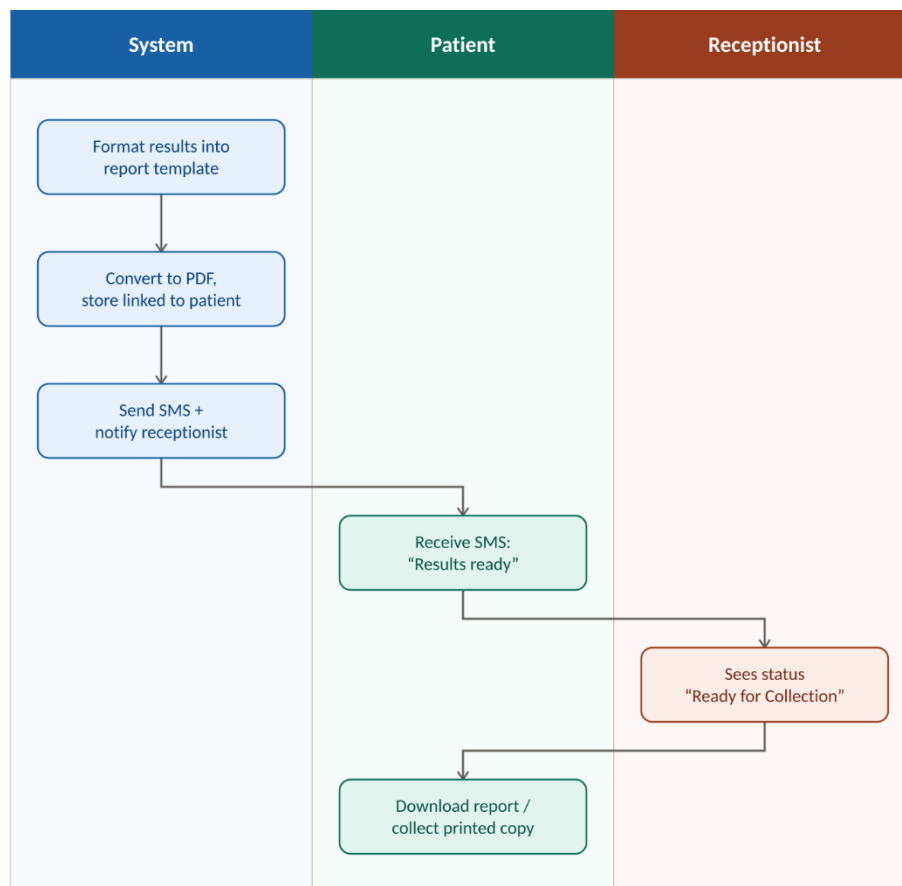


Key steps explained:

1. **Login and Queue View** - The technician logs in and views only the samples pending in their lab section, optionally filtered by test type, matching the "Filter by test type" step in the Lab Technician Workflow.
2. **Manual Result Entry** - The technician reads values directly from the equipment display/printout and enters them into the system.
3. **Automatic Flagging** - As each value is entered, the system checks it against the acceptable range for that test and automatically applies an H (high) or L (low) flag where relevant, rather than relying on the technician to remember reference ranges.
4. **Save and Trigger Downstream Actions** - Once results are saved, the sample status changes to "Result Ready", which automatically triggers report generation and notification (see 2.3.2.4) rather than requiring a separate manual step.
5. **Repeat / End** - The technician continues with the next pending sample or logs out at the end of the shift.

Report Generation & Notification

Figure 16: Activity diagram - Report Generation & Notification



Key steps explained:

1. **Report Formatting** - Once a technician saves results, the system automatically formats them into the standard report template.
2. **PDF Conversion and Storage** - The report is converted to PDF and stored in the database, linked to the patient's record for later retrieval.
3. **Notification** - An SMS is sent to the patient informing them the results are ready, and the receptionist's dashboard is simultaneously updated, reflecting the "Notify the customer through system" step from the Lab Technician Workflow and the "Send SMS to patient" step from the Overall System Flow.
4. **Retrieval** - The patient can either log into the patient portal to view/download the PDF report, or collect a printed copy physically from the receptionist, matching the two end-paths shown in the Patient Workflow diagram ("Access/Download Report" vs "Collect Results Physically").

2.4 Interface Design

Interface design defines how users interact with the Laboratory Information System. This section presents the screen layout principles, design rationale, and the key interface templates that form the foundation of all screens in the system. The interface design prioritises usability, consistency, and accessibility across all user roles.

2.4.1 Screen Layout Principles

The Laboratory Information System follows a consistent screen layout across all user interfaces to ensure familiarity and ease of use. The layout incorporates the following design elements:

Navigation Structure

All screens within the system use a sidebar navigation menu on the left side of the interface. The sidebar displays the system logo at the top, followed by role-specific navigation links. The navigation items are organised hierarchically, with the most frequently used options placed at the top. The current active page is highlighted using a blue background indicator to provide clear visual feedback to the user.

Content Area

The main content area occupies the right portion of the screen. It includes a header bar displaying the current page title and breadcrumb navigation. The content is organised using card-based layouts, where related information and controls are grouped into visually distinct cards with subtle borders and background shading.

Colour Scheme

The interface employs a professional medical colour palette. The primary colour is a calming blue (#2563EB) used for primary buttons, active navigation states, and important highlights. The background uses a light grey (#F8FAFC) to reduce eye strain during extended use. Text is primarily dark grey (#1E293B) for readability. Success states use green, warnings use amber, and errors use red.

Typography

The system uses the Inter font family for all interface elements. Headings use a semi-bold weight at 24px for page titles and 18px for section headings. Body text uses a regular weight at 14px. Form labels use a medium weight at 12px with uppercase lettering and letter spacing for clarity.

Form Design

All forms follow a consistent layout with labels positioned above input fields. Required fields are marked with an asterisk. Input fields use rounded corners and subtle borders that change colour on focus. Form validation messages appear below the relevant field in red text. Primary action buttons are positioned at the bottom of forms, aligned to the right.

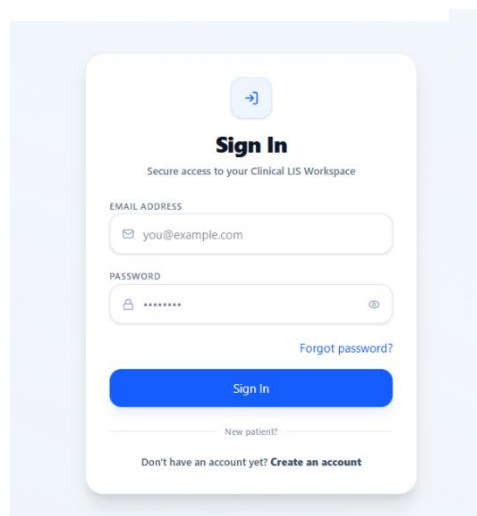
2.4.2 Key Interface Screens

The following screens represent the primary interfaces of the Laboratory Information System, demonstrating the application of the layout principles described above.

Login Screen

The login screen provides a clean, centred interface for user authentication. It features the system logo, email address and password fields, a primary sign-in button, and links for password recovery and account creation. The interface is designed to be simple and distraction-free, focusing the user's attention on the authentication task.

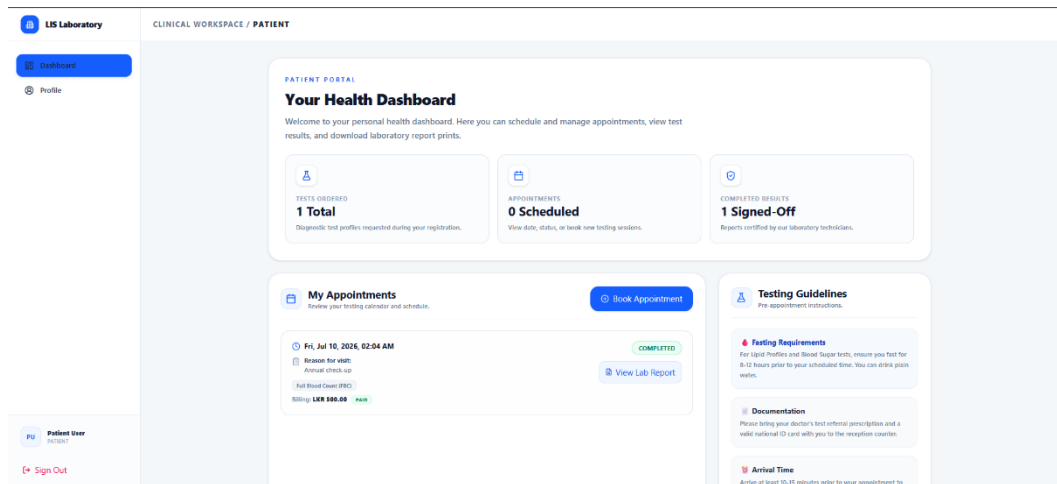
Figure 17: Login screen



Patient Dashboard

The patient dashboard provides a comprehensive overview of the patient's laboratory activities. It displays summary statistics for tests ordered, appointments scheduled, and completed results. The interface includes a My Appointments section showing upcoming and past appointments with status indicators, a Book Appointment button, and testing guidelines for patient reference.

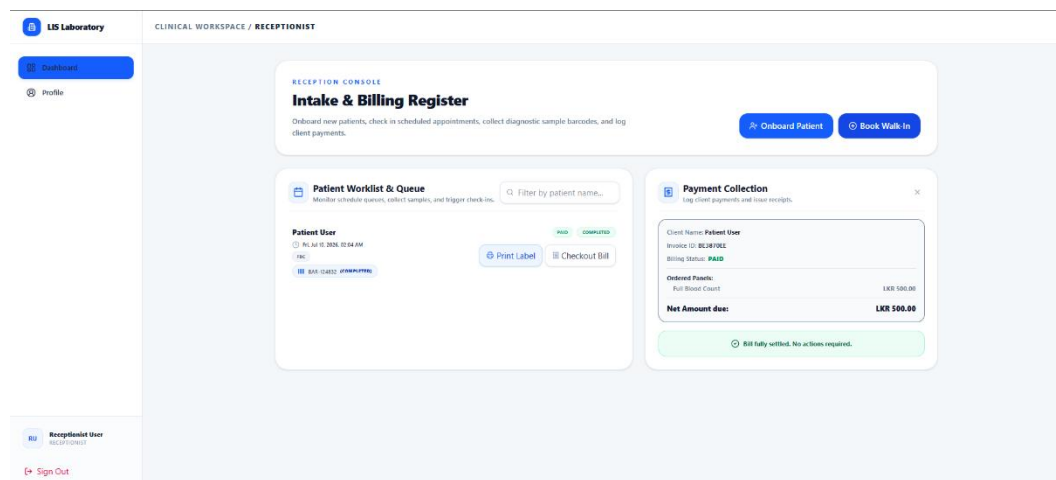
Figure 18: Patient dashboard



Receptionist Dashboard

The receptionist interface is designed for efficient patient intake and billing management. It features a patient worklist and queue showing scheduled appointments with their status, quick action buttons for onboarding patients and booking walk-in appointments, and a payment collection panel that displays billing details and allows payment processing.

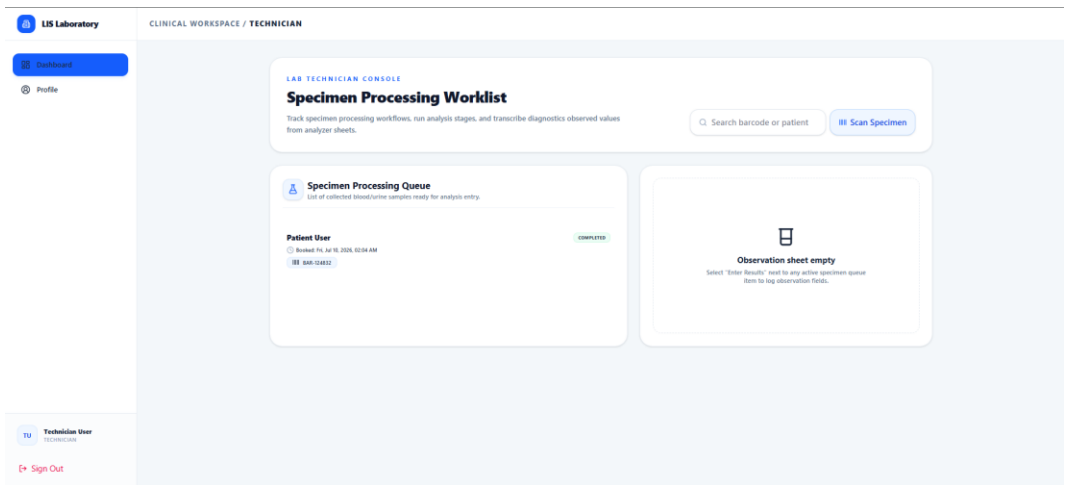
Figure 19: Receptionist dashboard & billing register



Technician Worklist

The technician interface focuses on specimen processing and result entry. It displays a specimen processing queue showing collected samples ready for analysis, with patient information and barcode identifiers. The observation sheet area provides structured input fields for entering test results. The interface is optimised for efficient data entry with minimal navigation.

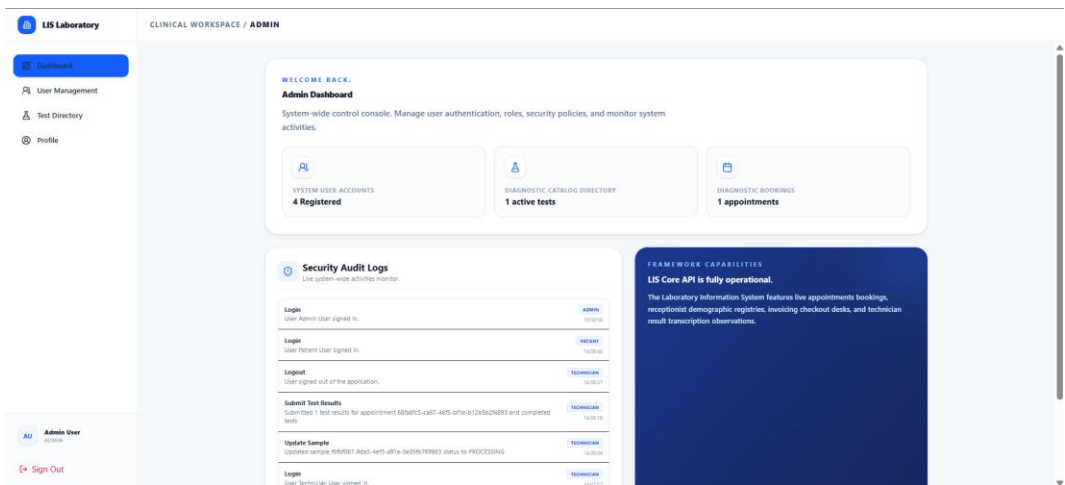
Figure 20: Technician test processing worklist



Administrator Dashboard

The admin dashboard serves as the system control centre. It displays summary statistics for registered users, active tests, and appointment counts. The security audit logs section shows recent system activities with timestamps. The interface provides access to user management, test directory management, and profile settings through the sidebar navigation.

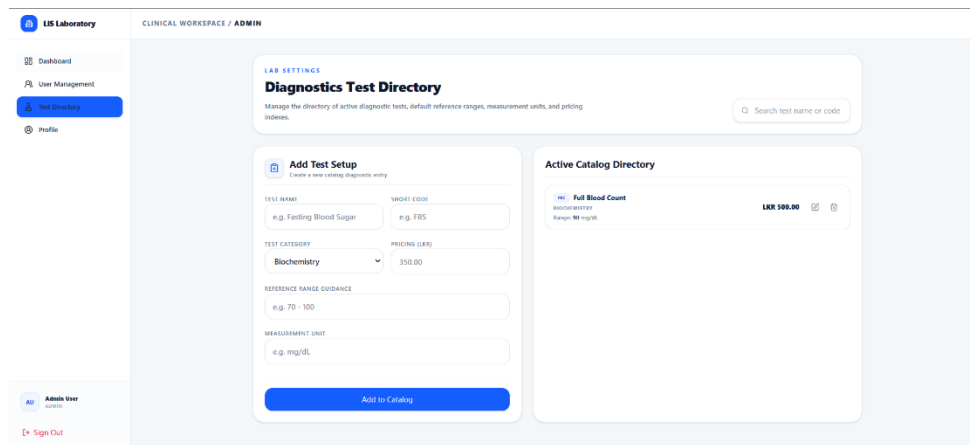
Figure 21: Administrator dashboard



Test Catalogue Management Flow

The test catalogue management process allows administrators to define and maintain the laboratory's test offerings. Each test entry includes a unique test code, descriptive name, medical category (e.g., Hematology), measurement unit, price, reference range values, and a detailed description. Tests can be marked as active or inactive in the catalogue.

Figure 22: Test catalogue interface

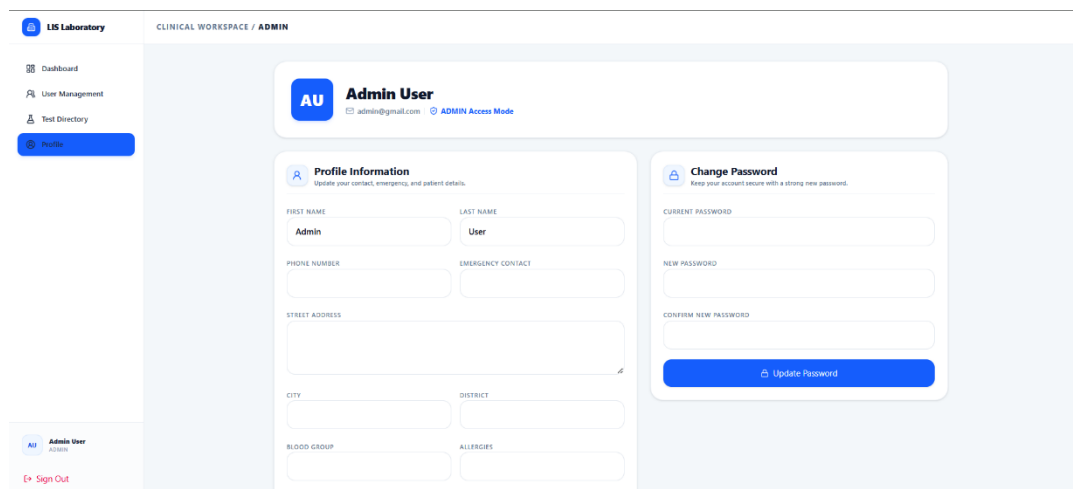


The screenshot shows the 'Diagnostics Test Directory' interface. On the left is a sidebar with navigation links: Dashboard, User Management, Test Directory (selected), and Profile. The main content area has a header 'LAB SETTINGS Diagnostics Test Directory' with a search bar. Below this is the 'Add Test Setup' form with fields for Test Name (e.g., Fasting Blood Sugar), Short Code (e.g., FBS), Test Category (Biochemistry), Pricing (USD) (\$50.00), Reference Range Guidance (e.g., 70 - 100), and Measurement Unit (e.g., mg/dL). An 'Add to Catalog' button is at the bottom. To the right is the 'Active Catalog Directory' showing a list of tests, with 'Full Blood Count' (BIOCHEMISTRY, Range: 90 mg/dL, LAR 500.00) as an example.

Profile Management Flow

All users can access their profile management interface to update personal information and change their account password. The profile page displays editable fields for contact information and medical details, alongside a secure password change form that requires the current password for verification.

Figure 23: Profile management



The screenshot shows the 'Profile Management' interface for 'Admin User'. The sidebar is the same as in Figure 22. The main content area has a header 'Admin User' with email 'admin@gmail.com' and 'ADMIN Access Mode'. Below this are two main sections: 'Profile Information' and 'Change Password'. The 'Profile Information' section has fields for First Name (Admin), Last Name (User), Phone Number, Emergency Contact, Street Address, City, District, Blood Group, and Allergies. The 'Change Password' section has fields for Current Password, New Password, and Confirm New Password, with an 'Update Password' button at the bottom.

2.4.3 Design Justification

The interface design decisions were made based on the following considerations:

- i. **Consistency** - All screens share common layout elements, colour schemes, and interaction patterns. This reduces the learning curve for users who interact with multiple parts of the system.
- ii. **Efficiency** - The card-based layout and sidebar navigation minimise the number of clicks required to access common functions. Technicians can enter results quickly, and receptionists can process patients efficiently.
- iii. **Accessibility** - The interface uses sufficient colour contrast for readability, clear typography, and intuitive navigation. Form validation provides immediate feedback to prevent errors.
- iv. **Responsiveness** - The layout adapts to different screen sizes using responsive design techniques, ensuring usability across desktop computers, tablets, and mobile devices.
- v. **Role-Based Adaptation** - Each user role sees only the navigation items and functions relevant to their responsibilities, reducing complexity and potential for errors.

Chapter 3: Hardware and Software Selection

The successful development and implementation of the proposed Laboratory Information System (LIS) require appropriate hardware and software resources. Selecting suitable development tools enables efficient system design, coding, testing, and maintenance, while appropriate implementation equipment ensures that the completed system operates reliably within the laboratory environment. This section describes the hardware and software required during both the development and implementation phases, together with the justification for each selection.

3.1 Development Hardware and Software Equipment

Development of hardware and software refer to the resources used by the development team during the analysis, design, coding, testing, and deployment of the Laboratory Information System.

3.1.1 Development Hardware

Developer Workstation (Laptop/Desktop Computer)

The primary hardware required for developing the Laboratory Information System is a laptop or desktop computer with sufficient processing power, memory, and storage capacity to support modern web development tools.

Table 15: Recommended specifications

Component	Specification
Processor	Intel Core i5 (10th Generation or above) / AMD Ryzen 5 or above
RAM	Minimum 8 GB (16 GB Recommended)
Storage	512 GB SSD
Display	Full HD (1920 × 1080)
Internet	Broadband Internet Connection

The development process requires running several software applications simultaneously, including Visual Studio Code, Node.js, PostgreSQL, web browsers, and testing tools. A multi-core processor and sufficient RAM improve compilation speed and multitasking performance, while an SSD provides faster project loading, database access, and application execution. A stable internet connection is also essential for downloading software packages, collaborating through GitHub, and accessing online documentation.

3.1.2 Development Software

1. Operating System - Windows 11

Windows 11 provides the development environment for building and testing the Laboratory Information System. It supports all required development tools and offers a stable platform for modern web application development. Windows 11 is compatible with Node.js, PostgreSQL, Visual Studio Code, and other software used throughout the project. Its wide hardware compatibility and regular security updates provide a reliable environment for software development.

2. Visual Studio Code

Visual Studio Code is used as the primary integrated development environment (IDE) for writing, editing, debugging, and managing the source code of the application. Visual Studio Code supports JavaScript, React.js, Node.js, HTML, CSS, SQL, and JSON. It provides intelligent code completion, debugging features, Git integration, syntax highlighting, and extension support, improving developer productivity and code quality.

3. React.js

React.js is used to develop the presentation layer of the Laboratory Information System. React.js enables the development of reusable user interface components, resulting in a responsive, interactive, and maintainable web application. It also improves user experience by updating only the necessary components without reloading the entire page.

4. Tailwind CSS

Tailwind CSS is used for designing the application's user interface. Tailwind CSS provides utility-based styling that simplifies interface development, promotes consistent design, and supports responsive layouts for different screen sizes.

5. Node.js with Express.js

Node.js and Express.js are used to implement the backend server and business logic of the system. Node.js efficiently processes multiple user requests using asynchronous programming, while Express.js simplifies API development, routing, authentication, and communication between the frontend and the database.

6. PostgreSQL

PostgreSQL serves as the relational database management system for the project. PostgreSQL provides secure and reliable storage for patient records, appointments, laboratory test results, billing information, and user accounts. It supports data integrity, scalability, and efficient query processing.

7. Postman

Postman is used to test and validate RESTful APIs developed for the backend. Postman enables developers to verify API functionality independently of the frontend, helping identify errors early and improving overall software quality.

8. Git and GitHub

Git is used for version control, while GitHub serves as the online repository for project collaboration. Version control allows developers to track source code changes, restore previous versions when necessary, and collaborate effectively within the development team.

9. Figma

Figma is used to design the graphical user interface before implementation. Figma allows developers to create interface prototypes, evaluate usability, and obtain feedback before coding begins, reducing design modifications during development.

3.2 Implementation Hardware and Software Equipment

Implementation hardware and software refer to the equipment required by the laboratory to successfully operate the completed Laboratory Information System.

3.2.1 Implementation Hardware

1. Desktop Computer or Laptop

The Laboratory Information System is accessed through a standard web browser; therefore, administrators, receptionists, and laboratory technicians require a desktop computer or laptop with moderate hardware specifications.

Table 16: Recommended specifications

Component	Specification
Processor	Intel Core i3 (10th Generation or above) / AMD Ryzen 3 or above
RAM	Minimum 8 GB
Storage	256 GB SSD
Display	Full HD Monitor
Internet	Broadband Internet Connection

The system is browser-based and does not require high-performance hardware. A standard office computer provides sufficient performance for patient registration, appointment management, laboratory result entry, report generation, and billing activities.

2. Laser Printer

A laser printer is used to print laboratory reports, invoices, and administrative documents. Although reports can be accessed electronically, many patients and healthcare providers still require printed copies. Laser printers provide fast, high-quality printing with low operating costs.

3. Barcode Printer

A barcode printer is used to print barcode labels for laboratory samples. Barcode labels improve sample identification accuracy, reduce manual errors, and simplify sample tracking throughout the laboratory testing process.

4. Barcode Scanner

A barcode scanner is used to scan sample labels during laboratory operations. Scanning barcodes enables quick retrieval of patient and sample information while minimizing manual data entry errors and improving operational efficiency.

5. Uninterruptible Power Supply (UPS)

A UPS supplies temporary backup power during unexpected electricity failures. Power interruptions may result in incomplete transactions or data loss. A UPS allows users to save ongoing work safely and shut down systems properly, improving system reliability.

6. Internet Router

A reliable internet connection is required for accessing the web application and transmitting data. The router enables communication between client computers and the application server while supporting online notifications and remote system access.

3.2.2 Implementation Software

1. Windows 10/11

The client computers operate using Windows 10 or Windows 11. These operating systems provide compatibility with modern web browsers and support all software required to operate the Laboratory Information System.

2. Google Chrome or Microsoft Edge

Modern web browsers provide access to the Laboratory Information System. The application is developed as a web-based system and is optimized for browsers that support current web technologies, responsive design, and secure HTTPS communication.

3. Node.js Runtime Environment

The backend server is deployed using the Node.js runtime environment. Node.js executes server-side application logic, processes user requests, and manages communication between the frontend and the PostgreSQL database.

4. PostgreSQL Database Server

The PostgreSQL database stores all system information permanently. The database maintains patient records, appointments, laboratory test results, reports, billing information, user accounts, and audit logs while ensuring data consistency, integrity, and security.

5. Email/SMS Notification Service

The notification service sends automated messages to patients. Patients receive notifications when laboratory reports become available, improving communication and reducing unnecessary visits to the laboratory.